

# Commerce XI Maths App

## Activities 3.1–3.2

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 3 - Complex Numbers | Total questions: 21

## Question 1 - Activity 3.1 · Step 1

If  $i = \sqrt{-1}$ , find the value of  $i^6$

### Options

- A. -1
- B. 1
- C.  $i$
- D.  $-i$

**Correct answer: -1**

### Explanation and steps

This is the stated step in solving Activity 3.1. Use  $i^2 = -1$ , equate real and imaginary parts when required, and simplify to obtain -1.

## Question 2 - Activity 3.1 · Step 2

If  $i = \sqrt{-1}$ , find the value of  $i^3$

### Options

- A.  $-i$
- B.  $i$
- C. 1
- D. -1

**Correct answer:  $-i$**

### Explanation and steps

This is the stated step in solving Activity 3.1. Use  $i^2 = -1$ , equate real and imaginary parts when required, and simplify to obtain  $-i$ .

## Question 3 - Activity 3.1 · Step 3

If  $i^6 = -1$ , find the coefficient of  $y$  after simplifying  $-15i^6y$

### Options

- A. 15
- B. -15
- C.  $15i$
- D.  $-15i$

**Correct answer: 15**

### Explanation and steps

This is the stated step in solving Activity 3.1. Use  $i^2 = -1$ , equate real and imaginary parts when required, and simplify to obtain 15.

## Question 4 - Activity 3.1 · Step 4

If  $i^3 = -i$ , simplify  $i^3(y + 4)$

### Options

- A.  $-(y + 4)i$
- B.  $(y + 4)i$
- C.  $-i(y - 4)$
- D.  $y + 4$

**Correct answer:**  $-(y + 4)i$

### Explanation and steps

This is the stated step in solving Activity 3.1. Use  $i^2 = -1$ , equate real and imaginary parts when required, and simplify to obtain  $-(y + 4)i$ .

## Question 5 - Activity 3.1 · Step 5

Simplify the equation

$$x + 2i = 7x - 15i^6y + i^3(y + 4),$$

using  $i^6 = -1$  and  $i^3 = -i$

### Options

- A.  $x - 15y + 2i = 7x - (y + 4)i$
- B.  $x + 15y + 2i = 7x + (y + 4)i$
- C.  $x - 15y + 2i = 7x + (y + 4)i$
- D.  $x + 15y + 2i = 7x - (y + 4)i$

**Correct answer:**  $x - 15y + 2i = 7x - (y + 4)i$

### Explanation and steps

This is the stated step in solving Activity 3.1. Use  $i^2 = -1$ , equate real and imaginary parts when required, and simplify to obtain  $x - 15y + 2i = 7x - (y + 4)i$ .

## Question 6 - Activity 3.1 · Step 6

In  $x - 15y + 2i = 7x - (y + 4)i$ ,  
the coefficient of  $i$ , on the left side is

### Options

- A. 2
- B. -2
- C. 0
- D.  $i$

**Correct answer:** 2

### Explanation and steps

This is the stated step in solving Activity 3.1. Use  $i^2 = -1$ , equate real and imaginary parts when required, and simplify to obtain 2.

## Question 7 - Activity 3.1 · Step 7

Equating the real parts of  $x - 15y + 2i = 7x - (y + 4)i$  gives

### Options

- A.  $-6x - 15y = 0$
- B.  $-6x + 15y = 0$
- C.  $6x - 15y = 2$
- D.  $-6x - 15y = 2$

**Correct answer:**  $-6x - 15y = 0$

### Explanation and steps

This is the stated step in solving Activity 3.1. Use  $i^2 = -1$ , equate real and imaginary parts when required, and simplify to obtain  $-6x - 15y = 0$ .

## Question 8 - Activity 3.1 · Step 8

Equating the imaginary parts of  $x - 15y + 2i = 7x - (y + 4)i$  gives

### Options

- A.  $2 = -(y + 4)$
- B.  $-2 = -(y + 4)$
- C.  $2 = y + 4$
- D.  $0 = -(y + 4)$

**Correct answer:**  $2 = -(y + 4)$

### Explanation and steps

This is the stated step in solving Activity 3.1. Use  $i^2 = -1$ , equate real and imaginary parts when required, and simplify to obtain  $2 = -(y + 4)$ .

## Question 9 - Activity 3.1 · Step 9

Simplify the equation  $x - 15y = 7x$

### Options

- A.  $-6x - 15y = 0$
- B.  $-6x + 15y = 0$
- C.  $6x - 15y = 0$
- D.  $-6x - 15y = 2$

**Correct answer:**  $-6x - 15y = 0$

### Explanation and steps

This is the stated step in solving Activity 3.1. Use  $i^2 = -1$ , equate real and imaginary parts when required, and simplify to obtain  $-6x - 15y = 0$ .

## Question 10 - Activity 3.1 · Step 10

Solve  $2 = -(y + 4)$  for  $y$

**Options**

- A. -6
- B. 6
- C. -2
- D. 2

**Correct answer: -6**

**Explanation and steps**

This is the stated step in solving Activity 3.1. Use  $i^2 = -1$ , equate real and imaginary parts when required, and simplify to obtain -6.

## Question 11 - Activity 3.1 · Step 11

If  $y = -6$  and  $-6x - 15y = 0$ , find  $x$

**Options**

- A. 15
- B. -15
- C. 9
- D. 6

**Correct answer: 15**

**Explanation and steps**

This is the stated step in solving Activity 3.1. Use  $i^2 = -1$ , equate real and imaginary parts when required, and simplify to obtain 15.

## Question 12 - Activity 3.1 · Step 12

Solve the equations  $-6x - 15y = 0$  and  $2 = -(y + 4)$ . The ordered pair  $(x, y)$  is

**Options**

- A. (15, -6)
- B. (-15, 6)
- C. (9, -6)
- D. (15, 6)

**Correct answer: (15, -6)**

**Explanation and steps**

This is the stated step in solving Activity 3.1. Use  $i^2 = -1$ , equate real and imaginary parts when required, and simplify to obtain (15, -6).

### Question 13 - Activity 3.1 · Step 13

If  $x = 15$  and  $y = -6$ , find  $x + y$

**Options**

- A. 9
- B. -9
- C. 21
- D. -21

**Correct answer: 9**

**Explanation and steps**

This is the stated step in solving Activity 3.1. Use  $i^2 = -1$ , equate real and imaginary parts when required, and simplify to obtain 9.

### Question 14 - Activity 3.2 · Step 1

If  $w^3 = 1$ , which factor multiplied by  $c + aw + bw^2$  gives  $a + bw + cw^2$ ?

**Options**

- A.  $w^2$
- B.  $w$
- C.  $w^3$
- D.  $(1)/(w^2)$

**Correct answer:  $w^2$**

**Explanation and steps**

Use  $w^3 = 1$  together with  $1/w = w^2$  and  $1/w^2 = w$ . Substitution in this step gives  $w^2$ .

### Question 15 - Activity 3.2 · Step 2

Complete the equivalent fractions:  $(a)/(w^2) = (a \text{ ___})/(w^3)$

**Options**

- A.  $w$
- B.  $w^2$
- C. 1
- D.  $-w$

**Correct answer:  $w$**

**Explanation and steps**

Use  $w^3 = 1$  together with  $1/w = w^2$  and  $1/w^2 = w$ . Substitution in this step gives  $w$ .

### Question 16 - Activity 3.2 · Step 3

Complete the equivalent fractions:  $(b)/(w) = (b \text{ \_\_\_})/(w^3)$

**Options**

- A.  $w^2$
- B.  $w$
- C. 1
- D.  $-w^2$

**Correct answer:  $w^2$**

**Explanation and steps**

Use  $w^3 = 1$  together with  $1/w = w^2$  and  $1/w^2 = w$ . Substitution in this step gives  $w^2$ .

### Question 17 - Activity 3.2 · Step 4

If  $w^3 = 1$ , simplify  $(a)/(w^2)$

**Options**

- A.  $aw$
- B.  $aw^2$
- C.  $a$
- D.  $(a)/(w)$

**Correct answer:  $aw$**

**Explanation and steps**

Use  $w^3 = 1$  together with  $1/w = w^2$  and  $1/w^2 = w$ . Substitution in this step gives  $aw$ .

### Question 18 - Activity 3.2 · Step 5

If  $w^3 = 1$ , simplify  $(b)/(w)$

**Options**

- A.  $bw^2$
- B.  $bw$
- C.  $b$
- D.  $(b)/(w^2)$

**Correct answer:  $bw^2$**

**Explanation and steps**

Use  $w^3 = 1$  together with  $1/w = w^2$  and  $1/w^2 = w$ . Substitution in this step gives  $bw^2$ .

## Question 19 - Activity 3.2 · Step 6

If  $w^3 = 1$ , simplify  $(a)/(w^2) + (b)/(w) + c$

### Options

- A.  $aw + bw^2 + c$
- B.  $aw^2 + bw + c$
- C.  $a + b + c$
- D.  $a + bw + cw^2$

**Correct answer:  $aw + bw^2 + c$**

### Explanation and steps

Use  $w^3 = 1$  together with  $1/w = w^2$  and  $1/w^2 = w$ . Substitution in this step gives  $aw + bw^2 + c$ .

## Question 20 - Activity 3.2 · Step 7

In the fraction  $(aw + bw^2 + c)/(c + aw + bw^2)$ , the numerator and denominator are

### Options

- A. equal
- B. additive inverses
- C. conjugates
- D. reciprocals

**Correct answer: equal**

### Explanation and steps

Use  $w^3 = 1$  together with  $1/w = w^2$  and  $1/w^2 = w$ . Substitution in this step gives equal.

## Question 21 - Activity 3.2 · Step 8

If  $w^3 = 1$ , find  $(a + bw + cw^2)/(c + aw + bw^2)$  in terms of  $w^2$

### Options

- A.  $w^2$
- B.  $w$
- C. 1
- D.  $-w^2$

**Correct answer:  $w^2$**

### Explanation and steps

Use  $w^3 = 1$  together with  $1/w = w^2$  and  $1/w^2 = w$ . Substitution in this step gives  $w^2$ .